STRUCTURED DATA

Structured data is very organized, has a definite shape and format, and fits predefined data type patterns. It follows a defined schema and is easily searchable because it is well organized. The data in a database is considered structured data because before it is stored in the database, it is expected to follow a strict schema. For example, data representing dates, numbers, or Booleans must be in a specific format.

Queries against structured data return results in exact matches. That is, we are concerned with finding documents that match the search criteria—but not how well they match. Such search results are binary: we either have a result or have no result—there's no "maybe" result. For example, it doesn't make sense to expect "maybe" canceled flights when searching for "canceled flights in the last month." There may be zero or more, but the search should not return "close match to canceled flights" results.

We do not worry about *how well documents match*—just that they match at all. Hence, no relevancy score (a positive number given to each result that indicates how closely the result matches the query) is attached to the results. A traditional database search is this sort: fetch all the flights that were canceled in the last month, weekly bestseller books, the activity of a logged-in user, and so on.

> **DEFINITION** *Relevancy* refers to the degree to which a search engine's results match a user's query. It's a mechanism for indicating how closely the results match the original query. The search engine uses relevance algorithms to determine which documents are closely related to the user's query (that is, how relevant they are) and produces a positive number called a *relevancy score* for each result based on how closely the result matches the query. The next time you search on Google, look closely at the search results: the top results are very closely related to what you are looking for, and hence we can say they are *more relevant* than the bottom entries in the list of results. Google internally assigns a relevancy score to each result and most likely sorts them by this score: the higher the score, the more relevant the result, and hence the more likely it is to be at the top of the page.

UNSTRUCTURED DATA

Unstructured data, on the other hand, is unorganized and follows no schema or format. It has no predefined structure. Unstructured data is the bread and butter of most modern search engines. Examples include blog posts, research articles, emails, PDFs, audio and video files, and so on.

> **NOTE** In addition to structured and unstructured data, there's another category: semi-structured data. This data falls pretty much in between structured and unstructured data. It is nothing but unstructured data with some metadata describing it.

For unstructured data, Elasticsearch offers full-text search capabilities that allow us to search for specific terms or phrases within large amounts of unstructured text. Full-text (unstructured) queries try to find results that are relevant to the query. That is, Elasticsearch searches all the documents that are most suited for the query. For example, if a